

Kangwei Li

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About Me

I am currently pursuing a Bachelor of Science in Mathematics at Xi'an Jiaotong University in Xi'an, China. I intend to pursue a Ph.D. in Statistics or Operations Research upon completion of my undergraduate studies. My research interests include probability theory and duality in optimization.

Education

Harvard University

Jan 2026 – Present

Visiting Undergraduate Student Program

Cambridge, MA

- Courses: *Stochastic Process, Convex Optimization, Decision Theory, Probability II*

Xi'an Jiaotong University

Sept 2023 - Present

Honors Science Program in Mathematics

Xi'an, China

- GPA: 4.0/4.3

- Courses: *Real Analysis, Probability, Mathematical Statistics, Functional Analysis, Complex Analysis*

Hangzhou High School & Xi'an Jiaotong University

Sept 2021 – Jun 2023

Honors Youth Program (Special Class for the Gifted Young)

Hangzhou & Xi'an, China

Research Experience

Research on Wasserstein Policy Gradient Method

Mar 2026 - Present

Advised by Prof. Rui Gao, University of Texas at Austin

- Proved a strong duality theorem for constrained MDPs with proper infinite dimensional state space and action space.
- Applied Wasserstein policy gradient as a primal-dual approach toward constrained MDPs.

Biologically plausible Neural Networks

Jun 2025 - Sept 2025

Supervised by Prof. Jianyong Sun, Xi'an Jiaotong University

- Reviewed the shortcomings of backward algorithm
- Investigated biologically plausible neural network models, including equilibrium propagation
- Explored potential improvements to backpropagation from a biologically grounded perspective

Selected Awards & Honors

First Prize, The 16th Chinese Mathematics Competition(CMC)

Nov 2024

China Mathematical Association

China National Scholarship (0.25% winning rate nationally)

Dec 2024

Xi'an Jiaotong University

Experience

AI for Mathematics Seminar

Guangzhou, China

Peking University

Jan 2025

- Learned Lean, a mathematical proof assistant and functional programming language.
- Finished a formalization of a theorem in convex optimization theory with Lean as a group project.

Skills

Programming Languages: Python, R, Lean

Language: English (Proficient), Mandarin (Native)